

20-7-26

3.

Breast Cancer Classification Performance

Find False Negative

Dataset = 1800 samples

Total malignant samples = 1800

Test data = 20%

Test samples per class:

$$1800 \times 20\% = 360$$

Given:

$$TP = 210$$

$$FN = ?$$

$$TP + FN = 360$$

$$FN = 360 - 210 = 150$$

Calculate Performance

Given:

$$TP = 210$$

$$TN = 190$$

$$FP = 30$$

$$FN = 150$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{210 + 190}{210 + 190 + 30 + 150} = \frac{400}{580}$$

$$\text{Accuracy} = 0.6897$$

$$\text{Accuracy} = 68.97\%$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{210}{210+30} = \frac{210}{240} \\ = 0.875 = 87.5\%$$

$$\text{Sensitivity} = \frac{TP}{TP+FN} = \frac{210}{210+50} = \frac{210}{360} \\ = 0.5833 = 58.33\%$$

$$\text{Specificity} = \frac{TN}{TN+FP} = \frac{190}{190+30} = \frac{190}{220} \\ = 0.8636 = 86.36\%$$

Effectiveness in Medical Diagnosis:

The classifier has high precision (87.5%), meaning most cases predicted as malignant are actually malignant. It also has good specificity (86.36%), including it correct identifier benign cases. The low sensitivity (58.33%) which is a serious concern in breast cancer diagnosis. Improving sensitivity is important to reduce missed cancer cases.

4.

Model Performance Analysis

i) Identify the epoch where overfitting begins

Epoch	Training Loss	Validation Loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

The model begins to overfit at 4

ii) 1. Early Stopping:

* Stop training when the validation loss starts increasing to prevent further overfitting.

2. Dropout Regulation:

* Randomly deactivate some neurons during training, reducing the model's dependence on specific features.

iii) Effect of overfitting on generalization

Overfitting occurs when the model learns the training data too closely, including noise and irrelevant patterns. As a result, it performs very well on the training data but poorly on unseen test data. This reduces the model's ability to generalize, leading to lower prediction accuracy and unreliable performance in real-world applications.